Subba Reddy Oota

Curriculum Vitae

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Education

2020– **Ph.D in Neuro-computational models of language comprehension and production grounded in robots**, *Inria-Research, Bordeaux, France*, Advisors: Prof. Alexder Frederic, Dr. Xavior Hinaut.

2016–2020 **Research Assistant in Cognitive Science**, *IIIT Hyderabad*, Advisors: Prof. Bapi S. Raju.

2014–2016 **M.Tech in Computer Science & Information Security**, *IIIT Hyderabad*, *CGPA - 8.4/10* .

2007–2011 **B.Tech in CSE**, *Koneru Lakshaiah College Of Engineering*, Guntur, *CGPA - 8.2/10*.

Publications

Mar 2020 – **Graph Convolutional Networks using Multiple Heat Kernels for Learning Structural to Functional Brain Mapping**, *(MICCAI-2021)*, In Review.

Feb 2020 – **Telugu Clickbait Detection: Dataset, Embeddings, Models and Anaysis of the Results**, *(IJCNN-2021)*, In Review.

Oct 2020 – **Unsupervised Graph based Telugu News Articles Text Summarization**, *(WiML@NeurIPS-2020)*, Accepted.

Oct 2020 – **Multi-Task Text Classification using Graph Convolutional Neural Networks for Resource-Poor Language**, *(WiML@NeurIPS-2020)*, Accepted.

Oct 2020 – **Healtech: A Deep Multi-Stage Method for Patient Hospitalization Risk Assessment**, *(WACV-2021)*, Accepted.

Jun 2021 – **Wound and Episode Level Readmission Risk or Weeks to Readmit: Why do patients get readmitted? How Long Does It Take for a patient to get readmitted?**, *(CODS-COMAD 2021)*, Accepted.

May 2020 – **Am I a Resource-Poor Language? Datasets, Embeddings, Models and Analysis for four different NLP tasks in Telugu Language**, *(Computational Linguistics Journal)*, In Review.

May 2020 – **Expert2Coder: Capturing Divergent Brain Regions Using Mixture of Regression Experts**, *(IEEE Transactions on Neural Networks & Learning Systems Journal)*, Asked for Revision.

Sep 2019 – **A Deep Multi-Modal Method for Patient Wound Healing Assessment**, *(Medical Imaging Meets NeurIPS Workshop)*, Accepted.

Jun 2019 – **Evaluating the Combination of Word Embeddings with Mixture of Experts and Cascading gcForest In Identifying Sentiment Polarity**, *(KDD Wisdom Workshop-2019)*.

May 2019 – **Towards Automated Evaluation of Handwritten Assessments**, *Vijay Rowtula, Subba Reddy Oota, C.V Jawahar*, (ICDAR-2019).

Mar 2019 – **StepEnCog: A Convolutional LSTM Autoencoder for Near-Perfect fMRI Encoding**, *Subba Reddy Oota, Vijay Rowtula, Manish Gupta, Raju S. Bapi*, (IJCNN-2019).

Nov 2018 – **Mixture of Regression Experts in fMRI Encoding**, *Subba Reddy Oota, Adithya Avvaru, Naresh Manwani, Raju S. Bapi, (NeurPS-2018 ViGIL workshop)*.

Oct 2018 – **A Deep Autoencoder for Near-Perfect fMRI Encoding**, *Subba Reddy Oota, Vijay Rowtula, Manish Gupta, Raju S. Bapi,(NeurPS-2018 Spatio Temporal Workshop)*.

Jul 2018 – **Affect in Tweets using Experts Model**, *Subba Reddy Oota, Adithya Avvaru, Mounika Reddy Marreddy, Radhika Mamidi, (PACLIC 2018)*.

Jun 2018 – **fMRI Semantic Category Decoding using Linguistic Encoding of Word-Embeddings**, *Subba Reddy Oota, Naresh Manwani, Raju S. Bapi,(ICONIP 2018)*.

Nov 2017 – **fMRI Semantic Category Decoding using Linguistic Encoding of Word2Vec**, *Subba Reddy Oota, Naresh Manwani, Raju S. Bapi,(ViGIL@NIPS 2017).*

Sep 2017 – **Believe It or Not! Identifying Bizarre News in Online News Media**, *Vijaysaradhi Indurthi, Subba Reddy Oota, Manish Gupta, Vasudev Varma,(CODS-COMAD 2017).*

Aug 2017 – **Clickbait detection using word embeddings**, *Vijaysaradhi Indurthi, Subba Reddy Oota, (Clickbait Challenge 2017).*

June 2017 – **Multi-ARM Active Transfer Learning for Telugu Sentiment Analysis**, *Subba Reddy Oota, Vijay Rowtula, Radhika Mamidi, (IAL @ ECML-PKDD 2017).*

Mar 2017 – **Tag me a Label with Multi-ARM: Active Learning for Telugu Sentiment Analysis**, *Subba Reddy Oota, Sandeep Sricharan Mukku, Radhika Mamidi, (DaWak-2017).*

Feb 2017 – **Fermi at SemEval-2016 Task 7: Detection and Interpretation of Homographic puns in English Language**, *Vijaysaradhi Indurthi, Subba Reddy Oota, (SemEval@ACL 2017).*

Dec 2016 – **Metastability of Cortical BOLD Signals in Maturation and Senescence**, *Shruti Naik, Subba Reddy Oota, Dipanjan Roy, Raju S. Bapi, (IJCNN-2017).*

Oct 2016 – **Relevance Detection and Argumentation Mining in Medical Domain**, *Vijaysaradhi Indurthi, Subba Reddy Oota, (Fire-Forum for Information Retrieval Evaluation,2016)*, Best Paper in CHIS track.

Work Experience

Jun 2016 – Current **Research Assistant**, IIIT Hyderabad.
I am currently working on various research projects such as Brain Decoding and Encoding using Mixture of LSTM Experts at Machine Learning and Cognitive Science Lab under the guidance of Dr. Naresh Manwani and Prof. Bapi S Raju, Graph Diffusion Kernels Using Functional Harmonics on Task Based fMRI under the guidance of Dr. Avinash Sharma and Prof. Bapi S Raju, Brain Encoding on Music Cognition fMRI data using Bidirectional LSTM under the guidance of Dr. Manish Gupta from Microsoft and Prof. Bapi S Raju.

Aug 2018 – **Sr. Data Scientist**, Woundtech Innovative Healthcare Solutions, Hyderabad.

Jun 2016 – Jul 2018 **Machine Learning Scientist**, TeradataAster, Hyderabad.
Research and develop Machine Learning models for text datasets received from the customers. Worked on word-vector representations such as Word2Vec, GloVe etc and topic modeling on some of the real world datasets and implemented these techniques for Aster. Worked on KNIME enhancements for TeradataAster. Worked on Aster analytics to verify how different classifiers like SVM, DT, Neural Networks, Linear Regression working on asterR with standard datasets and also compared with open source libraries. Worked on text analytics to apply NER, POS tagging, Sentiment Analysis and graph analytics like Page Ranking, Propagating the labels, different graph measures, HMM based models, PCA etc.

Apr 2016 – May 2016 **Data Scientist**, *Internship*, Fabulyst, Hyderabad.
Implemented a Naive Bayes algorithm to find the cross entropy probabilities of user personnel attributes to predict whether a user like a dress or not.

Jan 2016 – May 2016 **Teaching Assistant**, *Statistical Methods in Artificial Intelligence.*, IIIT Hyderabad, Mentoring 9 teams to apply Machine Learning techniques on problems related to Text Mining, Image Processing and NLP.

Aug 2015 – Dec 2015 **Teaching Assistant**, *Discrete Maths & Algorithms*, IIIT Hyderabad.

Sep 2011 – Jul 2014 **Project Engineer**, CDAC R&D, Mumbai.
Roles: Case Based Reasoning Algorithm design (CBR),Expert System Design, Database design, Java web development, Handling SMSPush Service, SMSPull Services, IVRS services etc.

Industry Projects

SQL-Map Reduce functions for Machine Learning, Datamining and Statistics

Python, R, SQLMR, Java

Aster Analytics is based on a Massively Parallel Processing (MPP) architecture, where tasks are run simultaneously across multiple nodes for efficient processing. The analytic functions are performed directly in the database to eliminate data movement and to leverage the MPP. The ability to use different Machine Learning techniques in the SQL queries enables data scientists to seamlessly use them in the queries without worrying about the internal implementation and allows them to iterate over models rapidly. I am involved in validating the Aster SQLMR functions in Machine Learning, Data Mining, Graph Analytics and Statistics. I am also involved in implementing KNIME enhancements.

IASF-Intelligent Advisory System for Farmers

Java,CBR based,Rule based engine

Intelligent Advisory System For Farmers was implemented based on Case Based Reasoning, where all the farmer queries store in database as cases, if a query have already solution exist, when same type query fired, system only generate the solution.

Achievements

PC Member **Program Committee member for the conference CODS-COMAD 2021**.

Webchair **Web Area Chair for the conference CODS-COMAD 2021**.

Reviewer **Reviewer for the conferences ICML-2021, CHIL-2021, WACV-2021, ICLR-2021, NeurIPS-2020, CODS-COMAD 2021, CHIL-2020, ML4H @NeurIPS-2019, ICON-2018, ICON-2017**.

Volunteer **Volunteer for the conferences NeurIPS-2020, ICML-2020, ICLR-2020** .

Meta-Reviewer **Meta-Reviewer for the workshop ML4H @ NeurIPS-2020**.

FDP **Given two lectures on CNN and Its Applications in Faculty Development Program at Central University Rajasthan, 2020**.

FDP **Given lecture on CNN and Its Applications in Faculty Development Program at NIT Warangal, 2020**.

Subreviewer **Subreviewer for the journal Frontiers in Neuroscience, and conferences EMNLP-2020, IJCNN-2020, IJCNN-2019, ICON-2019, ACCS-2019, IJCNN-2018** .

FDP **Given 1-hour lecture on Deep Learning Applications at MJCET, Hyderabad, 2020.**.

FDP **Given 4-days lecture on Machine Learning & Deep Learning in Faculty Development Program at SRKR Engineering college, Bheemavaram, 2019**.

Awards **Received best employee award at Woundtech Innovative Health Care Solutions, Hyderabad**.

Session Chair **Session chair for the conference ICONIP-2018**.

Awards **Received best paper award in CHIS track at FIRE-2016 from Springer**.

Awards **Received best performance from Xerox research in CHIS track at FIRE-2016**.

Awards Deans List-2 in M.tech CSIS 2nd Semester.

Gate Secured 98.9% and 1704 AIR in GATE-2014 Examination in CSE.

Showcase Winner of North-Eastern award, Finalists of mBillionth in domain of e-Agriculture for IASF project.

Relevant Coursework

Machine Learning	Digital Image Processing	Discrete Maths & Algorithms
Topics in ML	Advances in Data Mining	Artificial Neural Networks
Statistical Methods in AI	Computer Vision	Data Mining
Information Retrieval & Extraction	Advanced Problem Solving	Database Systems

Technical Skills

Languages	C, C++, Java, Python, MATLAB, R
Machine/Deep Learning Tools	Auto-Sklearn, Scikit Learn, Keras, Torch, NLTK, Spacy, Theano, Tensorflow, PyTorch, Azure ML Studio